

Standard algebraic manipulations show that the following polarizations hold for $n > 0$:

$$\langle \Omega, x\Phi_n \rangle = \sum_{k=0}^3 \frac{i^{-k}}{2} \phi_{k,n}(x) \quad \text{and} \quad \langle \Phi_n, y\Omega \rangle = \sum_{k=0}^3 \frac{i^k}{2} \phi_{k,n}(y)$$

For $n > 0$, the terms $\langle \Omega, x\Phi_n \rangle \langle \Phi_n, y\Omega \rangle$ and $\langle \Omega, y\Phi_n \rangle \langle \Phi_n, x\Omega \rangle$ appearing in θ_δ can then be written as a linear sum of positive functionals as follows:

$$\begin{aligned} \langle \Omega, x\Phi_n \rangle \langle \Phi_n, y\Omega \rangle &= \frac{1}{4} \sum_{k,m=0}^3 \phi_{k,n}(x) \cdot i^{(+m)} \phi_{k+m,n}(y), \\ \langle \Omega, y\Phi_n \rangle \langle \Phi_n, x\Omega \rangle &= \frac{1}{4} \sum_{k,m=0}^3 \phi_{k,n}(x) \cdot i^{(-m)} \phi_{k+m,n}(y). \end{aligned}$$

And hence:

$$\langle \Omega, x\Phi_n \rangle \langle \Phi_n, y\Omega \rangle + \langle \Omega, y\Phi_n \rangle \langle \Phi_n, x\Omega \rangle = \frac{1}{2} \sum_{k=0}^3 \phi_{k,n}(x) \cdot (\phi_{k,n}(y) - \phi_{k+2,n}(y)).$$

In Equation (4), the above terms show up in θ_δ multiplied by $f_\delta(l_n)$. Aside from the value $l_n = 0$ for which we know $f_\delta(0) = 1/2$, each $f_\delta(l_n)$ might be positive or negative (as we noted, we can and do take a real f).

We then just need to be cautious about the sign of $f_\delta(l_n)$. Thus, we define:

$$a_\delta(k) := \begin{cases} 1 & \text{if } f_\delta(k) > 0 \\ 0 & \text{otherwise} \end{cases} \quad b_\delta(k) := \begin{cases} 1 & \text{if } f_\delta(k) < 0 \\ 0 & \text{otherwise} \end{cases}$$

Then, for each n at most one of the indices $a_\delta(l_n)$ and $b_\delta(l_n)$ is 1, and it holds that $f_\delta(k) = (a_\delta(k) - b_\delta(k)) \cdot |f_\delta(k)|$. Summing all terms, we obtain

$$\begin{aligned} \theta_\delta &= \underbrace{\omega \otimes \omega + \sum_{n>0} \sum_{k=0}^3 \frac{|f_\delta(l_n)|}{2} \phi_{k,n} \otimes (a_\delta(l_n)\phi_{k,n} + b_\delta(l_n)\phi_{k+2,n})}_{=: \theta_{\delta,+}} \\ &\quad - \underbrace{\sum_{n>0} \sum_{k=0}^3 \frac{|f_\delta(l_n)|}{2} \phi_{k,n} \otimes (a_\delta(l_n)\phi_{k+2,n} + b_\delta(l_n)\phi_{k,n})}_{=: \theta_{\delta,-}} \end{aligned}$$

Hence we get the desired decomposition $\theta_\delta = \theta_{\delta,+} - \theta_{\delta,-}$ with $\theta_{\delta,\pm}$ defined as

$$\begin{aligned} \theta_{\delta,+} &:= \omega \otimes \omega + \sum_{n>0} \sum_{l=0}^3 \frac{|f_\delta(l_n)|}{2} \phi_{l,n} \otimes (a_\delta(l_n)\phi_{l,n} + b_\delta(l_n)\phi_{l+2,n}). \\ \theta_{\delta,-} &:= \sum_{n>0} \sum_{l=0}^3 \frac{|f_\delta(l_n)|}{2} \phi_{l,n} \otimes (a_\delta(l_n)\phi_{l+2,n} + b_\delta(l_n)\phi_{l,n}). \end{aligned} \quad (5)$$

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With the definition of Eq. (5) in hands, we now focus on estimating the entropy $S_1(\theta_{\delta,+}/\|\theta_{\delta,+}\|)$.

Define τ_δ as the positive functional $\theta_{\delta,+}$ restricted to the first tensor component. It is then expressed as follows:

$$\tau_\delta(x) := \theta_{\delta,+}(x \otimes \mathbb{1}) = \omega(x) + \sum_{n>0} \sum_{k=0}^3 \frac{|f_\delta(l_n)|}{2} \phi_{k,n}(x) \quad \text{for } x \in \mathcal{B}(\mathcal{H}). \quad (6)$$

This decomposition of τ_δ into pure states $\phi_{k,n}$ is indeed convergent in norm, because we assume that the net satisfies the condition 2.10(1), i.e., $\dim(\mathcal{H}_N)$ grows bounded by an almost exponential function $C \exp(N^\kappa)$ for some $C > 0$ and $\kappa \in (0, 1)$.

The above decomposition into vector states enables us to invoke Corollary 2.3. We first note that, since the operator $\theta_{\delta,+}$ is not normalized, so is τ_δ not normalized. One can calculate its norm $C_\delta := \|\tau_\delta\| = \|\theta_{\delta,+}\| = \theta_{\delta,+}(\mathbb{1} \otimes \mathbb{1})$ as

$$C_\delta = \sum_{N \geq 0} 2 \dim(\mathcal{H}_N) |f_\delta(N)|,$$

because every factor $|f_\delta(N)|/2$ appears $4 \dim(\mathcal{H}_N)$ -times, where the factor 4 is due to the sum in k . Likewise, with the fact that all $\phi_{k,n}$ are pure states, the

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